

Cert to Competent

A coaching fundamentals primer — the frameworks that turn a qualification into real, defensible programming.

WHO THIS IS FOR

Newer coaches and PTs who've got the certificate and want the judgement to go with it — the "why" behind the programming, not just another rep scheme to copy.

FROM THE COACH

I've been coaching for 20 years, and I've watched more good athletes get held back by lazy programming than by bad genetics. A certificate teaches you the movements. It doesn't teach you why one plan works and another one just fills an hour.

This primer is deliberately short, I don't believe in padding it out to look impressive. Six chapters, each one a framework you can start applying in your very next session — not a literature review. I've already done the digging through the current research; what's here is the distilled, decided version, with a brief "receipt" in each chapter if you want to see where it comes from.

Read it once end to end. Then keep it on hand, because you'll use it more as a reference than as reading.

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Why Periodisation Beats Random Programming

Training is a stressor. Selye's General Adaptation Syndrome describes what happens next: your body absorbs the stress and adapts — or, if the stress never lets up, breaks down under it instead. Bannister's Fitness-Fatigue model puts a number on the same idea: **Performance = Fitness – Fatigue**. Every session adds both. Fatigue just clears faster.

Random programming has no mechanism for either side of that equation. It doesn't systematically increase the stimulus once an athlete adapts, and it has no plan for shedding fatigue before it buries the fitness sitting underneath it. Periodisation is simply the stress-recovery cycle, applied on purpose instead of left to chance.

It also explains why the *order* you build qualities in matters. Issurin's block periodisation model rests on "residual training effects" — how long an adaptation sticks around once you stop directly training it. Maximal strength and aerobic base decay slowest. Power, rate of force development, and reactive strength decay fastest. So the sequence that works: build the long-lasting qualities first, convert them into more specific strength qualities, then peak the fast-decaying ones right before they're needed.

THE RECEIPT

Williams et al. (2017, *Sports Medicine*) found periodised training outperforms non-periodised/constant training for strength gains. A separate review (Grgic et al.) found the periodisation *model* — linear vs undulating — barely moves hypertrophy once volume is equated. The win isn't the model. It's having a plan at all.

SHORT TERM

Fatigue managed, no staleness.

MEDIUM TERM

Peaking timed to competition.

LONG TERM

Fewer injuries, longer careers.

DON'T MAKE THIS MISTAKE

Periodising gets an athlete MORE from training, not less. A planned deload isn't a day off — it's where fatigue clears so the fitness underneath it can show up. Skipping deloads, or adding in extra work because the athlete "feels good," stacks fatigue nobody is tracking. That's the line between functional overreaching (planned, short-term, pays off) and non-functional overreaching (unplanned, weeks to dig out of) — per the ECSS/ACSM joint consensus statement on overtraining (Meeusen et al., 2013). Feeling good mid-block isn't a green light.

How to Structure a Session

A session isn't a random list of exercises — it's ordered on purpose. The template: **Prep** → **Power** → **Main** → **Secondary** → **Accessories**.

Prep uses the RAMP method — Raise (elevate heart rate and tissue temperature), Activate (wake up the muscles about to be loaded), Mobilise (move the joints through the ranges the session needs), Potentiate (prime the nervous system with a lighter, faster version of what's coming).

Power comes next, while the nervous system is freshest — jumps, throws, olympic lift variations. These need maximum quality per rep, so they can never sit after your athlete is already fatigued.

Main is the primary strength lift for the day — squat, deadlift, bench, or the sport-specific equivalent. This is where load and technical demand are highest.

Secondary is a related but less demanding lift, building the same pattern at lower stakes.

Accessories close the session — this is where you attack the weak spots and build the strengths a needs analysis flagged, with sets, reps, and tempo picked for that individual, not copy-pasted from a template.

THE RECEIPT

NSCA exercise-order guidance: sequence fatigue-sensitive, high-skill work first. Ask an athlete to jump explosively after their legs are already smoked from squats, and you get neither a good jump nor a safe one.

REMEMBER: "IT DEPENDS"

The exact structure bends around the athlete in front of you — training age, sport, current phase of the season, injury history. A first-year athlete and a ten-year veteran don't need the same accessory work, even when the main lift on the page looks identical.

Exercise Selection: Stop Looking For "The Best"

I get asked "what's the best hamstring exercise" more than almost any other question, and my answer never changes: there isn't one. There is no universally "best" hamstring exercise, best core exercise, best anything. There's the right exercise for this goal, this person, in this session. Chasing size under load? Romanian deadlifts or Nordic curls. Training speed-specific hamstring function? Something loaded closer to sprint speed. The question was never "what's best" — it's "best for what."

That same needs-analysis thinking is what protects your athletes from injury. Take a session that strings together front squats, deadlifts, front rack lunges, and RDLs — four individually excellent exercises. Programmed back-to-back, three of them (deadlift, lunge, RDL) load the exact same tissue: the lumbar erectors and posterior chain, with no recovery window between bouts.

The injury mechanism isn't the exercise — it's fatigue-driven form breakdown. As the lower back fatigues through the session, trunk flexion under load increases, the athlete loses neutral spine, and load meant for muscle shifts onto the discs and passive tissue instead.

DON'T BE A LAZY COACH

Stacking spine-loading movement after spine-loading movement without accounting for cumulative fatigue isn't programming — it's guessing, and the lower back pays for it. Know what tissue you've already loaded before you load it again.

Understanding Strength Qualities

"Strength" isn't one thing — it sits on a spectrum, the force-velocity curve, and different qualities live at different points on it:

Maximal Strength — the most force you can produce, regardless of speed.

Strength-Speed — moving heavy loads as fast as possible.

Speed-Strength — moving light loads as fast as possible.

Power — force × velocity, the whole curve at once.

Rate of Force Development — how fast force is produced; decisive under ~250ms (a sprint start, a change of direction).

Starting Strength — force production from a dead stop.

Reactive / Elastic Strength — force through the stretch-shortening cycle (bounding, jumping).

Strength Endurance — sustaining force output across repeated efforts.

Eccentric Strength — controlling force as a muscle lengthens under load.

Even athletes with one obvious headline quality train more than that quality. A powerlifter's outcome is maximal strength, but rate of force development matters at the sticking points — off the chest in bench, off the floor in deadlift — which is why speed work with submaximal loads shows up in most serious powerlifting programs. Strength endurance matters too: an athlete needs the work capacity to handle the training volume that builds the strength in the first place.

THE TAKEAWAY

The qualities all serve the one goal. Training only 1RM attempts, and nothing else, is a fast way to plateau.

Deloads: Why Less (Sometimes) Is More

A deload isn't a reward, and it isn't optional once fatigue has been banked — it's the part of the plan where the adaptation an athlete has been training for actually shows up.

Volume is the primary lever, not intensity. Cut volume 25–45% for lower recovery needs, 40–60% for higher, while holding intensity closer to the athlete's working range. Crashing intensity down to 60–65% is the common mistake.

THE RECEIPT

Bell et al. (2025, *Strength and Conditioning Journal*), backed by the Bell et al. (2023) Delphi consensus, found maintaining intensity while cutting volume delivers superior results for strength and power outcomes than crashing intensity down.

WHY THIS MATTERS

Skip the deload, or quietly add in extra sets because an athlete "feels good," and you're stacking fatigue nobody is tracking. The research draws a clear line between functional overreaching (planned, short recovery, comes back as a gain) and non-functional overreaching (unplanned, weeks to months to dig out of) — the ECSS/ACSM joint consensus statement on overtraining (Meeusen et al., 2013). Feeling good mid-block isn't a green light. It's fatigue that hasn't been paid for yet.

Programming Smart, Not Busy

In 20 years of coaching, the biggest difference between a good session and a wasted hour was never the exercises on the page, it was the intent behind them. Every idea in this primer points at the same discipline: know why an exercise, a rep range, or a deload is in the program — or take it out.

A session full of exercises that "feel productive" but weren't chosen against a needs analysis isn't thorough. It's noise. Filling an hour isn't the job. Building an athlete is.

Before you write the next session, ask three questions:

- ✓ What quality is this developing?
- ✓ Has this tissue already been loaded today?
- ✓ Does this athlete need more volume, more intensity, or more recovery, right now?

If you can't answer those three, you're not programming yet — you're just filling space.

KEEP THIS ON HAND

Know the why behind every choice you make.

I put this primer together because I wish someone had handed it to me early in my own coaching career, it would've saved a lot of trial and error. Come back to it whenever a program starts feeling like a guess instead of a plan.

Build your programs around the athlete in front of you, not a template you found online. Twenty years in, that's still the whole job.

Thanks for reading, but moreover, thanks for becoming a better coach.

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